CAPSTONE PROJECT REPORT

(Project Term January-May 2025)

WEB-BASED MENTAL HEALTH SCREENING AND GUIDANCE SYSTEM

Submitted by

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Project Group Number: KC318

Course Code: CSE439

Under the Guidance of

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DECLARATION

We hereby declare that the project work entitled The Web-Based Mental Health Screening and Guidance Systemis an authentic record of our own work carried out as requirements of Capstone Project for the award of B.Tech degree in Computer Science and Engineering from Lovely Professional University, Phagwara, under the guidance of Chetna Viad Kwatra, during January to April 2025. All the information furnished in this capstone project report is based on our own intensive work and is genuine. Project Group Number: ……

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CERTIFICATE

This is to certify that the declaration statement made by this group of students is correct to the best of my knowledge and belief. They have completed this Capstone Project under my guidance and supervision. The present work is the result of their original investigation, effort and study. No part of the work has ever been submitted for any other degree at any University. The Capstone Project is fit for the submission and partial fulfillment of the conditions for the award of B.Tech degree in Computer Science and Engineering from Lovely Professional University, Phagwara.

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Acknowledgement

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TOPIC APPROVAL PERFORMA

School of Computer Science and Engineering (SCSE)

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PROPOSED TOPIC : The Web-Based Mental Health Screening and Guidance System

Qualitative Assessment of Proposed Topic by PAC		
Sr.No.	**Parameter**	**Rating (out of 10)**
1	Project Novelty: Potential of the project to create new knowledge	6.18
2	Project Feasibility: Project can be timely carried out in-house with low-cost and available resources in the University by the students.	6.45
3	Project Academic Inputs: Project topic is relevant and makes extensive use of academic inputs in UG program and serves as a culminating effort for core study area of the degree program.	6.18
4	Project Supervision: Project supervisor's is technically competent to guide students, resolve any issues, and impart necessary skills.	7.73
5	Social Applicability: Project work intends to solve a practical problem.	5.64
6	Future Scope: Project has potential to become basis of future research work, publication or patent.	7.00

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Final Topic Approved by PAC: **The Web-Based Mental Health Screening and Guidance System**

Overall Remarks: Approved

PAC CHAIRPERSON Name: 13897::Dr. Deepak Prashar **Approval Date:** 23 Apr 2025

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1. INTRODUCTION

With the advent of technology, society has started to suffer from mental health challenges like stress, anxiety, and depression, which affects people of all ages and professions. Despite the widespread increase in understanding mental health, most people still confront significant obstacles when trying to access help, including social stigma and lack of awareness.

This predicament highlights mental illness as one of the most under-resourced problems in public health that is a result of insufficient addressable services made available for technology alongside user privacy, control, and ongoing aid.To respond to the problems, we developed a holistic system to assist users in monitoring and enhancing their mental well-being.

Our project presents a MENTOR: WEB BASED MENTAL HEALTH SCREENING AND GUIDANCE SYSTEM that provides a private and safe platform for users to assess their mental health using validated clinically approved questionnaires like the PHQ-9 for depression and the GAD-7 for anxiety.

Support with assessment is provided through AI powered analytics that autonomously determine the level of symptoms and provide feedback to clients in real time. The system features advanced data storage that enables clients to hold and control their personal data.

Our approach enables the client to seek support as they wish and provides them with the ability to make changes anytime permitting clients with utmost self-sufficiency. Also as an aimed to go beyond diagnosis and monitoring features, the system offers individualized suggestions such as tailored self-help to meet the user’s needs.

1.1. SYSTEM OBJECTIVES

- Facilitate secure user identification and profile identification.
- Conduct mental health assessments using PHQ-9 and GAD-7.
- Analyze test outcomes using artificial intelligence.
- Provide tailored advice regarding mental well-being.
- Showcse dashboards allowing users to track their mental health journey.

1.2. SYSTEM ARCHITECTURE OVERVIEW

The Mental Health Web System is structured to ensure seamless communication between the frontend, backend, database, and AI components while maintaining security and scalability. Below is an overview of the architecture:

1.2.1. Frontend (React.js)

- Guarantees an adaptive interface that is simple to engage with.
- Enables registration, login, participation in assessments, satisfaction
- view, recommendations, and assessment tracking.
- Exchanges information with back-end systems through the use of REST APIs.

1.2.2. Backend (Node.js with Express.js / Flask)

- Controls the business logic processes and the API endpoints.
- Oversees user authentication via JWT (JSON Web Tokens).
- Handles the storage and processing of user profiles and test results.
- Works alongside AI/ML models for the analysis of tests.

1.2.3. AI/ML Integration (Logistic Regression Model)

- Classifies severity of depression and anxiety using PHQ-9 and GAD-7 test responses.
- Trained model deployed as a microservice or embedded in the backend.

1.2.4. Deployment (AWS / Firebase / DigitalOcean)

- Hosts the backend server, frontend client, and MongoDB database.
- Ensures high availability, auto-scaling, and security with HTTPS and firewall

settings.

1.2.5. Communication Flow

- User interacts with frontend (React.js).
- Frontend sends requests to backend APIs (Node.js/Flask).
- Backend processes requests, interacts with MongoDB, and performs AI analysis.
- Backend sends responses back to frontend for display.

This architecture ensures modularity, easy scaling, high security, and efficient performance, providing a reliable mental health support system to users.

2. PROFILE OF THE PROBLEM / RATIONALE & SCOPE OF THE STUDY (PROBLEM STATEMENT)

2.1. PROBLEM STATEMENT

The rising prevalence of mental health problems (i.e., depression, anxiety, stress), has heightened the demand for affordable, accessible, confidential mental health support. While digital health has grown in response to this demand, the vast majority of live digital health solutions available on the market tend to be priced too high, offer little personal care options, and do not incorporate user engagement tools (e.g., tracking personal progress). Consequently, there is an urgent need for an inclusive web-based platform with standardized mental health assessments, personalized treatment recommendations using artificial intelligence (AI) technology, self-monitoring abilities.

2.2. RATIONALE

Improving mental health at scale means making sure that people can locate and find help, regardless of costs, social barriers, or physical distance, to receive help when necessary. In this case, a digital platform offering an array of verified mental health assessments, tailored feedback, and evidence-based approaches assists in bridging the gap between their need for early detection and professional intervention. Individuals have the ability to leverage self-assessments and AI analytics, receiving autonomous, guided, early intervention assistance, and the ability to mitigate the burden on health systems, and optimize the health of society.

2.3. SCOPE OF THE STUDY

The system focuses on:

- Building secure user authentication and profile management.
- Providing structured mental health assessments (PHQ-9 for depression and GAD-7 for anxiety).
- Delivering AI-generated severity assessment and individualized resources.
- Providing users with a historical view of tests and enable tracking of their mental health over time.
- Ensuring privacy, scalability, and usability to facilitate widespread adoption.

The project sets the stage for the eventual integration of telehealth, other psychological assessments, and multilingual capabilities in order to widen its reach.

3. EXISTING SYSTEM

Currently, there are numerous mobile applications and web portals that assess mental health and provide self-help resources. Well-known applications like BetterHelp, Calm, and Talkspace offer both therapy and general wellness options. However, these applications also have significant downsides, including:

High Cost: Many mental health apps can only be accessed by available paid subscriptions or costly therapy sessions.

- **Limited Personalization:** Existing platforms offer generic feedback or suggestions that may not deeply reflect an individual's distinct mental health, based on a scientific assessment.
- **No Progress Tracking:** Most applications do not have a comprehensive dashboard for users to track their test results or developments in their mental health over time.
- **Data Privacy Concerns:** Peforming assessment testing within an existing mental health or wellness application creates risks for users, especially if there is no end-to-end encryption or strong security measures in place to protect sensitive mental health data.

Although these applications offer valuable services, they fail to deliver a completely secure, user-friendly, AI-generated, and affordable system that is available to many. The deficiencies in existing mental health and wellness applications reveal a strong need for a comprehensive, user-friendly, and personalized Mental Health Web System like the one presented in this proposal.

3.1. INTRODUCTION

The current landscape of mental health solutions is mainly facilitated through mobile applications and web portals, that provide therapy sessions, meditation exercises, and self-help written articles (the content of these articles vary in quality). Examples of mobile applications or web sites like: BetterHelp, Calm and Talkspace are great in terms of creating access to mental health services for everyone. Meanwhile, while these applications and/or web-sites do fill a gap, they are expensive, are not personalized to the individual, and do not have vital features such as ongoing self-monitoring and ongoing tracking of progress. As well, there are still considerable concerns regarding data safety and privacy. Although these applications/web-sites do serve as a search engine for people who want a certain level of attention to their mental health needs, there is a fundamental gap for free, AI-supported, highly personalized, and secure approach to user self-assessment, actionable AI insights into the user-self- assessment, and user self- monitoring.

3.2. CURRENT SOFTWARE

There are currently many mental health platforms on the market, providing online therapy sessions, guided meditations, stress relief exercises, or general wellness advice. Some of these include the following:

BetterHelp: An online counseling platform offering professional therapy sessions via chat, phone, or video.

- **Calm:** Mindfulness application that provides meditation, sleep help, and stress

relief.

- **Talkspace:** An online therapy platform that connects users with licensed therapists. While these services are helpful, they do have limitations:
- **Cost Barrier:** Professional therapy by these platforms can be expensive, limiting access to people with fewer financial resources.
- **Generalized Approach:** The advice and exercises are not tied to detailed or structured mental health assessments like PHQ-9 or GAD-7.
- **Lack of Self-Monitoring Tools:** The majority of these platforms do not give users the recorded opportunity to track their mental health assessment results over time.
- **Data Security Risks:** In some instances, protected health information is not considered sensitive or protected under any privacy law or standards or best-practices, limiting user trust.

3.3. WHAT'S NEW IN THE SYSTEM TO BE DEVELOPED

The purposed Mental Health Web System introduces several new and improved features compared to the current offertings:

- **AI-Driven Personalized Analysis:** Severity of mental health conditions is evaluated using scientifically validated logistic regression models.
- **Standardized Testing:** Requires validated tests (PHQ-9 and GAD-7), instead of a general self-assessement.
- **Secure and Private:** User authentication with JWT, secure database, and encrypted method of distributed storage.
- **Free and Accessible:** Platform is designed to be free or low-cost for ease of access to mental health for other people.
- **Progress Tracking:** Users can review their previous mental health assessments using a visual dashboard.
- **Resource Recommendations:** Users receive a curated list of self-help, professional help, and lifestyle recommendations based on the assessments.
- **Scalable and Responsive:** Modern technology stacks and services are cloud deployments friendly, and cloud scalable.

4. PROBLEM ANALYSIS

4.1. PRODUCT DEFINITION

The Mental Health Web System is a robust, AI mediated web system that aims to democratize mental well-being support. Users are able to self-assess their mental health using internationally-recognized scales (PHQ-9 for depression, GAD-7 for anxiety).

The platform allows people to securely authenticate, assess their mental health, utilize user tailored AI analysis and tracking files to offer discreet individual support on key trends, with actionable suggestions that range from self-help to referrals to professional support.

Unlike other apps that emphasize therapy with high subscription fees or professional involvement, this is about taking control of individual trajectories in a discreet and private manner for free at their own pace.

The modular and scalable nature of this platform allows us to integrate additional services in the future such as teletherapy, multiple languages, and community-based programs well-

User Authentication and Profile Management

- Secure sign-up/login using JWT authentication.
- Personal dashboard to manage assessments and view history.

Mental Health Assessments (PHQ-9, GAD-7)

- Standardized, clinically validated questionnaires.
- Instant scoring and categorization into severity levels.

AI/ML-Based Analysis

- Logistic Regression model classifies user scores.
- Dynamic, personalized interpretation of results.

Personalized Recommendations

- Tailored advice: lifestyle improvements, stress management, sleep hygiene.
- Curated articles, meditation guides, and crisis helpline suggestions.

Progress Tracking

- Visual dashboard showing trends over multiple assessments.
- Encourages self-awareness and early intervention.

Mobile-Responsive UI

- Accessible via desktop, tablet, and mobile devices.

Data Privacy and Security

- User data encrypted at rest and in transit.
- Confidentiality maintained through secure storage and strict authentication protocols.

Scalable Cloud Deployment

- Hosting on AWS, Firebase, or DigitalOcean.
- Easy to scale with growing user demand.

4.2. FEASIBILITY ANALYSIS:

Before developing the Mental Health Web System, a thorough feasibility analysis was performed, taking into consideration the technical, economic, and operational feasibility of the project. This analysis is important to know if the project is possible to build, if there are properly available resources, time, and technology.

4.2.1. TECHNICAL FEASIBILITY

The selected technologies for this project (React.js, Node.js/Flask, MongoDB, and Logistic Regression AI models) are very stable, freely available open-source technologies with plenty of support. These are being considered:

- Frontend: React.js provides an efficient and modular way to create responsive user interfaces
- Backend: Node.js (with express) or Flask can handle the API request and integrate with the AI models
- Database: MongoDB can store user profiles, test results, and resource data in a scalable and flexible way.
- Authentication: JWT tokens manage sessions in a secure manner • AI/ML: Logistic Regression is straightforward, interpretable, and will perform well on binary/multi-class mental health score classifications.
- Deployment: Cloud providers such as AWS or firebase have great hosting options for this project and provide ways for auto-scaling, HTTPS secure hosting, and offers qr code scanning.

4.2.2. ECONOMIC FEASIBILITY

The system is cheap for either a student-level or startup-level deployment:

- Most of the development technologies (React.js, Node.js, MongoDB, Flask, scikit-learn for ML) are either free and or open source.
- The cloud deployment can be done on either the AWS Free Tier, Firebase Hosting, or very low-cost VPS services like DigitalOcean.
- Start-up costs are essentially just a domain name (optional) and very compensable server fees.

- There are no expensive third-party APIs or licensing fees.

Overall, the project gives a great deal of value for the price and is even feasible for growth on a large scale in the future.

4.2.3. Operational Feasibility

The system is designed to be easily usable, maintainable and operate:

- The user interface is minimalistic and intuitive making the learning curve easy.
- Authentication is easy and is done securely and the user never sees the credentials.
- The dashboard provides easy visualizations of the user's mental health progress which encourages users to re-engage over time.
- Maintenance (i.e. testing criteria updates, etc., resource additions, or including other types of AI models and improving existing ones, etc.,) can be easily handled as a backend update while the system is running and usable.

In practice, the platform can run autonomously without human manually involved in operation on a "day-to-day" basis.

PROJECT PLAN

The Mental Health Web System is being developed in a methodical sequence of phases based on a weekly schedule. Each phase allows for incremental building, testing and integration of different components of the project.

Week 1–2: Research & Concept Development

Research mental health topics by looking at screening tools such as the PHQ-9 for depression and the GAD-7 for anxiety.

- Research existing mental health apps and services.
- Identify project goals, user requirements, and system requirements

Week 3: Technology Stack Finalization and UI/UX Design

Make decisions on the tech stack: React.js, Node.js (Express.js or Flask), MongoDB, JWT Authentication.

In addition to deciding on the stack, ensure UI wireframes are developed for (Sign-up/Login screen, Assessment screen, Dashboard).

Final architecture for frontend, backend and database connections are decided.

Week 4–6: Frontend and Backend Development

Build out the frontend components including forms for authentication, questionnaire pages for assessment, and dashboard for displaying results.

Develop backend APIs for user management, submissions of questionnaires and storage of results.

Implement the mongoDB database to store users profiles and mental health records.

Week 7: AI/ML Model Integration :

- Create and train a Logistic Regression model to work with our application.
- Integrate the model with the backend to predict a user's mental health level of risk, based on the questionnaire responses.
- Conduct tests of the predictive accuracy of the model using sample data.

Week 8: System Testing and Debugging

- Conduct Unit Testing of each individual module (Authentication, Start Assessment, Result analysis).
- Integration Testing to confirm the connection between the front end and the backend work together as expected to conduct the programming functions correctly.
- Identify and fixes bugs and optimizing the system.

Week 9: Deployment and Hosting

- Deploy the application on cloud services like **AWS**, **Firebase**, or **DigitalOcean**.
- Perform security checks and final optimization.
- Ensure mobile responsiveness of the web app.

Week 10: Final Presentation and Documentation

- Prepare detailed project report (architecture diagrams, flowcharts, screenshots).
- Create a user manual for using the system.
- Finalize and rehearse project demonstration for presentation day.

5. REQUIREMENT ANALYSIS

5.1. INTRODUCTION

The Mental Health Awareness and Early Guidance Web System is intended to help users self-assess their mental health and receive recommendations based on established assessment tools.

A web-based application, it incorporates some basic machine learning features to provide intelligent analysis and recommendations.

The requirements analysis stage includes both functional and non-functional aspects of the system. This stage makes sure that the design meets the needs of users, as well as the reliability and scalability needs of the system.

The purpose of the web app is to support secure access to customized mental health screening, AI-assisted interpretation of results , and curated resources — all via a user-friendly, responsive web app interface.

The system must:

- Ensure a secure and private authentication of users.
- Allow users to perform mental health assessments easily.
- Analyze the answers and estimate mental health risk levels using a Logistic Regression model.
- Provide tailored self-help and professional referrals.
- Allow users to track their progress over time.

The software system should be lightweight, responsive, and scalable to accommodate a growing number of users, while maintaining high standards of data privacy and security.

5.2. GENERAL DESCRIPTION

The Mental Health Awareness Web System functions as a full-stack web application which enables a self-examination of mental health, AI early risk prediction, and customizable

recommendations.

The entire system is modular and consists of several pieces working together.

5.2.1. User Authentication and Profile Management

- Sign-up/Login system that requires only an email address and password.
- Authentication is managed with JWT (JSON Web Tokens) protecting user sessions.
- For each user a profile is associated containing record or basic details (name, age, gender) and history of their assessment.

5.2.2. Mental Health Assessments Module

- Standardized questionnaires like PHQ-9 (for depression) and GAD-7 (for anxiety).
- User completes multiple-choice questionnaires (rating severity or frequency of symptoms).
- Submission scores are calculated automatically.

5.2.3. AI/ML Risk Analysis Module

- The Web System is trained a Logistic Regression model that predicts the probability of a user experiencing depression or anxiety based on their responses.
- This classification/distribution categorizes users into severity levels (Minimal, Mild, Moderate, Severe).
- This classification is helpful in creating appropriate recommendations for users.

5.2.4. Personalized Recommendations System

- Based on assessment scores and risk predictions:
    - Self Help Recommendations: articles, meditation guides, mental health exercises, etc.
    - Professional Help Recommendations: helpline numbers, local mental health professionals etc.
    - **Lifestyle Tips**: sleep management, diet, exercise routines to support mental well-being.

5.2.5. Progress Tracking and Dashboard

- The dashboard displays:
  - Previous test scores.
  - Risk levels over time (visualized through graphs).
  - Personalized advice history.
- Encourages users to monitor improvements and motivates continuous engagement.

5.2.6. System Admin Panel (Optional - Future Scope)

- Admins can manage users, update resources, and monitor system health.
- Analytics on the number of users, common issues, and average risk levels.

5.3. SPECIFIC REQUIREMENTS

The Specific Requirements describe precisely what the Mental Health Awareness Web System must accomplish, how it must respond, and which quality standards will be applied.

It is divided into **Functional Requirements** and **Non-Functional Requirements**.

5.3.1. Functional Requirements :

Module	Input	Process	Output
User Registration and Login	User's email and password	Validate credentials using secure authentication (JWT)	Successful login or error message
Profile Management	User details (age, gender, etc.)	Save and retrieve user profile data	Personalized dashboard
Mental Health Assessment	Questionnaire responses (PHQ-9, GAD-7)	Score responses and calculate total score	Display severity level (Minimal, Mild, Moderate, Severe)
Risk Prediction (AI Model)	Assessment scores	Predict mental health risk using Logistic Regression	Risk category (Low, Moderate, High)
Personalized Recommendations	Risk level	Fetch relevant self-help resources and professional contacts	List of recommendations
Progress Tracking	Historical assessment data	Analyze trends	Graphs and visual progress reports

5.3.2. Non-Functional Requirements

Category	Requirement
Performance	System must load the assessment form and dashboard in less than 3 seconds.
Security	Passwords must be hashed and user session must be protected through JWT authentication.
Scalability	Must support scaling up to thousands of users without significant down time to the system.
Usability	UI must be simple, responsive (desktop and mobile friendly).
Reliability	Must handle failed network requests gracefully and save user progress automatically.
Data Privacy	User mental health data needs to be encrypted, and staff must not be able to access it without the consent from the user.
Maintainability	Code base must be modularly built, and there must be enough code comments/documentation so that updates are manageable.
Availability	System uptime must exceed 99% during typical, active hours.
Portability	Web app must load in all major web browsers (Chrome, Edge, Firefox, Safari).

6. Design

6.1. System Architecture

The **Mental Health Awareness System** follows a **client-server architecture** with AI/ML integration at the backend for mental health risk prediction.
It includes the following layers:

1. **Frontend (Client Side)**

    - Built with **React.js**.
    - Allows users to register, log in, take assessments, view results, and access recommendations.
    - Communicates with backend APIs over secure HTTPS.

2. **Backend (Server Side)**

    - Developed with **Node.js (Express.js)** or **Flask (Python)**.
    - Handles user authentication (JWT), stores and fetches data, processes assessment scores.
    - Integrates an AI model (Logistic Regression) to predict mental health risks based on assessment results.

3. **Database**

- Uses **MongoDB** to store:
    - User profiles
    - Assessment histories
    - AI prediction results
    - Recommendation mappings

4. **Machine Learning Module**

    - Logistic Regression model trained on mental health datasets.
    - Hosted as part of backend or a separate lightweight microservice.
    - Predicts the user's mental health risk (Low, Moderate, High).

5. **Hosting/Deployment**

    - Application deployed on cloud platforms such as **AWS**, **Firebase**, or **DigitalOcean** for scalability and reliability.
        - User profiles
        - Assessment histories
        - AI prediction results
        - Recommendation mappings

6. **Machine Learning Module**

- Logistic Regression model trained on mental health datasets.
- Hosted as part of backend or a separate lightweight microservice.
- Predicts the user's mental health risk (Low, Moderate, High).

7. **Hosting/Deployment**

- Application deployed on cloud platforms such as **AWS**, **Firebase**, or **DigitalOcean** for scalability and reliability.

6.2. COMPONENTS OVERVIEW

Component	Role
Frontend (React.js)	Provides the user interface for assessments, progress tracking, and login system.
Backend (Node.js / Flask)	Handles API endpoints for registration, assessment submission, and AI model integration.
JWT Authentication	Secures user login and session management using token-based authentication.
MongoDB Database	Stores user data, assessment records, AI results, and recommendations securely.
AI/ML Module (Logistic Regression)	Predicts the user's mental health status based on PHQ-9 and GAD-7 inputs.

Deployment Server (AWS / Firebase / DigitalOcean)	Hosts the frontend, backend, and database services ensuring scalability and uptime.

6.3. SYSTEM DESIGN DIAGRAM

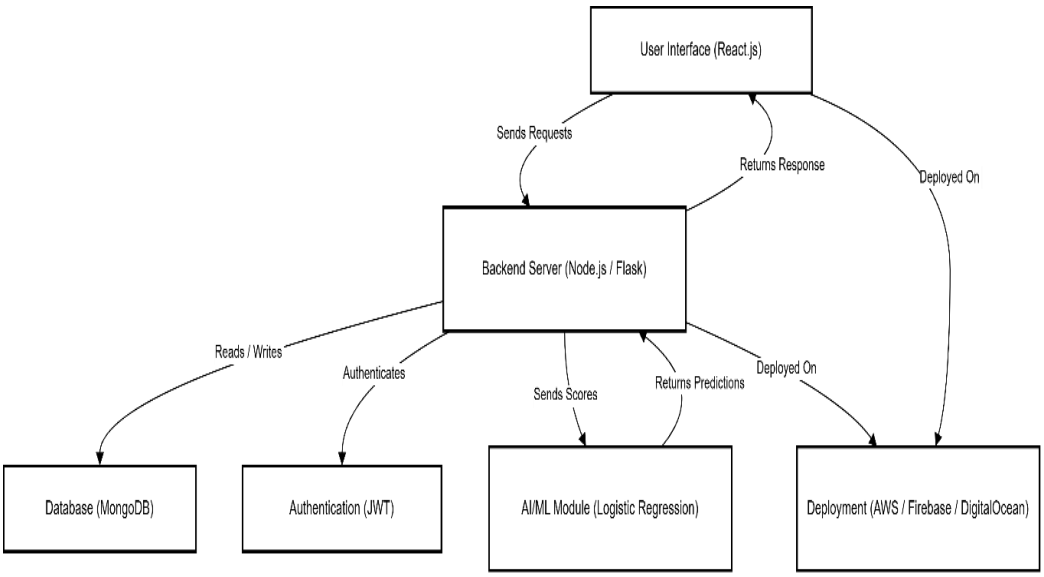


Fig-1 DFD Diagram Level 0

6.4. DATA FLOW DIAGRAM

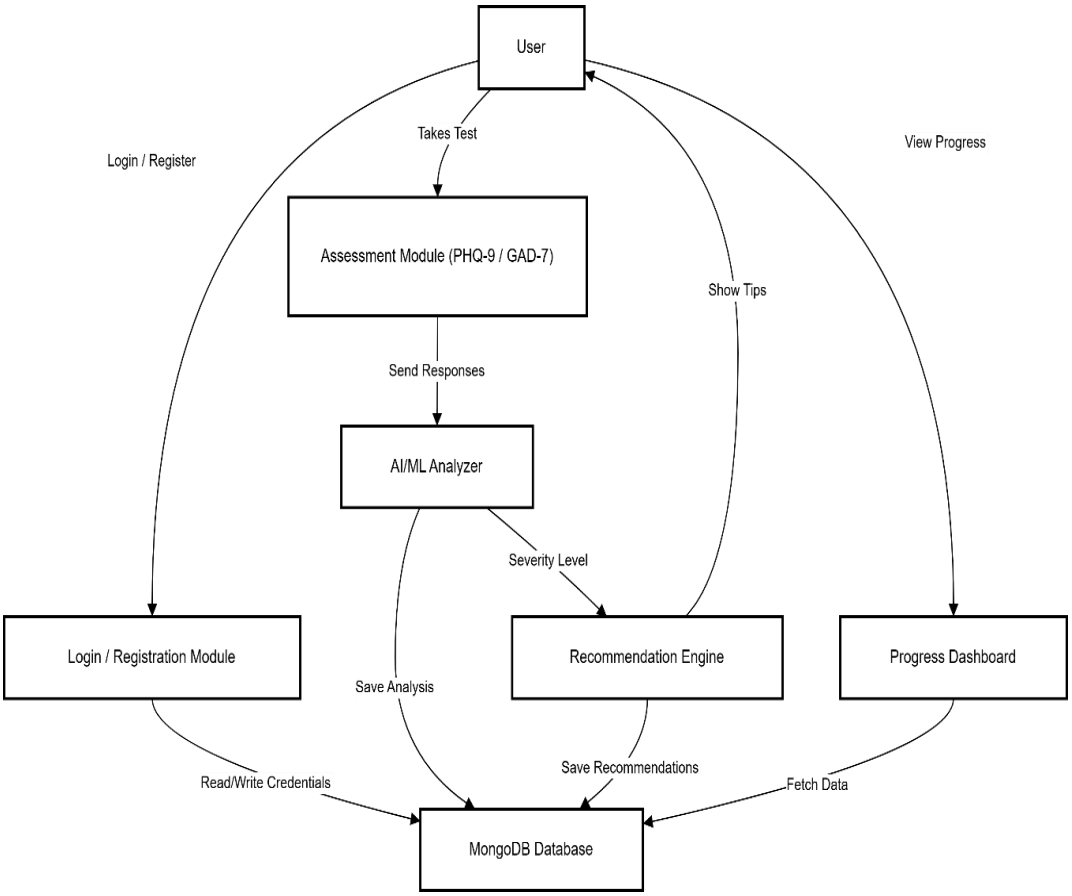


Fig- 2 DFD Diagram Level 1

6.5. Flow chart:

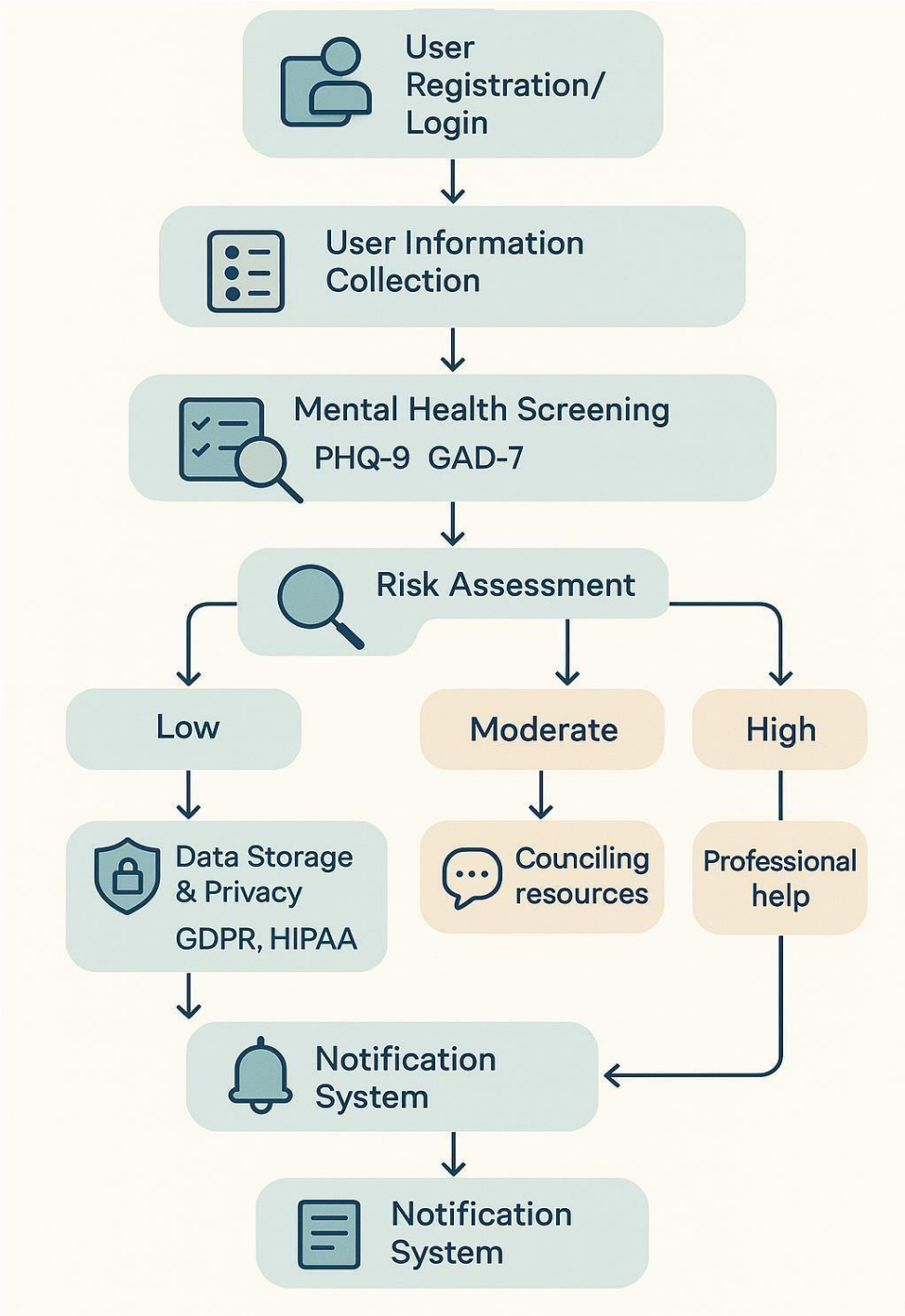


Fig- 3 Flow chart of the full project

6.5. Pseudo code

```
// Pseudo code for Mental Health Tracking and Support System


// Define Components
User Input: Mood, Symptoms, Stress Level, Sleep Quality, Activity Level
Database: Stores user history, validation, suggested resources
Feedback: Provides possible coping strategies, resources, connects potential professionals
Notifications: Reminders for check-in, exercises

// Initialization Phase
initialize system:
    Display "Welcome to the Mental Health Tracker"
    Set up user profile (age, gender, preferences, etc.)
    Set up database connection to record user input and progress history
    Set up notifications to remind user (e.g., daily check-in)

start system:
    Display "Please rate your current mental health status"
    Show options: [Mood, Stress Level, Symptoms, Sleep Quality, Activity Level]

// User Input Phase
start user check-in:
    Ask user to input their most current mood (scale of 1-10)
    Ask user to input their most current stress level (scale of 1-
    10) Ask user to rate their quality of sleep (scale of 1-10)
    Ask user to input any symptoms (e.g., anxiety, fatigue, sadness) Ask
    user to rate their activity level (scale of 1-10)
```

```
    Store all inputs in the database

// Analyze User Input and Provide Feedback analyze
user input:
    if mood is low or stress level is high:
        Display Its looks like you are stressed or down.Here are some relaxation techniques:"
        Recommendations: Exercises (deep breathing, meditation)
        Resources (links to articles, videos, or apps)
        If symptoms persist, recommend a mental health professional

    if low activity level:
        Display " You may wish to increase your level of physical activity to improve mental health. Here are some activity ideas:"
        Suggest light exercises or outdoor activities

    if sleep quality is low:
        Display " Low quality sleep can be detrimental to mental health. Here are some tips for improving both sleep quality and quantity:" Provide suggestions for good sleep hygiene.
        Suggest good sleep hygiene practices

// Regular Check-in Notifications
send check-in reminder:
    if it's time for a daily check-in:
 Notify user to log their mental health status for the day

// Track Progress
track user history:
```

```
    Retrieve previous check-ins from the database
        Generate progress chart (eg mood improvement over time)
        Create personalised feedback based on previous input

// Connect with Support
offer professional help:
    if user asks for professional help:
        Display a list of mental health professionals or helplines
        Provide option to book an appointment or contact emergency services


// System Shutdown
shutdown system:
    Display "Thank you for using the Mental Health Tracker"
    Save user data and progress
    Send reminders for the next check-in
```

7. TESTING

- Functional testing validates that the application behaves as necessary, meeting provided specifications, for all core features. Below is a proposed framework for functional testing for your mental health tracking system:

Test Plan Overview

- **Objective:** Verify that the core features of the app, such as user registration, daily check-ins, feedback mechanisms, progress tracking, and professional support, function correctly.
- **Scope:** All features related to user input, feedback, tracking, notifications, and integration with professional support systems.

Test Cases

1. **User Registration and Authentication**
    - **Test Case 1.1:** Verify user can register an account successfully.

        - **Precondition:** User is on the registration screen.
        - **Steps:**
            1. Enter valid username, email, and password.
            2. Click "Register."
        - **Expected Result:** User should be registered and logged into the system, redirected to the home screen.
    - **Test Case 1.2:** Verify login functionality with valid credentials.
        - **Precondition:** User has registered an account.
        - **Steps:**
            1. Enter valid username and password.
            2. Click "Login."
        - **Expected Result:** User should be logged in and redirected to the home screen.
    - **Test Case 1.3:** Verify login functionality with invalid credentials.
        - **Precondition:** User has an account.
        - **Steps:**
            1. Enter invalid username or password.
            2. Click "Login."
        - **Expected Result:** Error message should display, e.g., "Invalid credentials."

2. **Daily Check-In**
    - **Test Case 2.1:** Verify user can input mood, stress level, sleep quality, symptoms, and activity level.
        - **Precondition:** User is logged into the system.
        - **Steps:**
            1. Navigate to the daily check-in screen.
            2. Enter values for mood (scale 1-10), stress level (scale 1- 10), sleep quality (scale 1-10), symptoms, and activity level (scale 1-10).

        3. Click "Submit."
    - **Expected Result:** User input should be saved in the database, and a success message should appear.
  - **Test Case 2.2:** Verify that the system provides personalized feedback after the check-in.
    - **Precondition:** User has completed the check-in.
    - **Steps:**
        1. After submitting the check-in, the system analyzes the input.
        2. System should display tailored recommendations based on user input (e.g., stress-relief exercises if stress level is high).
    - **Expected Result:** Relevant coping strategies and resources should be displayed.

3. **Progress Tracking**
  - **Test Case 3.1:** Verify that the system tracks and displays mood improvement over time.
    - **Precondition:** User has logged multiple daily check-ins.
    - **Steps:**
        1. Navigate to the progress tracking section.
        2. View mood trends over time (displayed in a graph/chart).
    - **Expected Result:** The system should show a visual representation of mood trends based on previous inputs.

4. **Notifications**
  - **Test Case 4.1:** Verify that the system sends a reminder for the daily check-in.
    - **Precondition:** User has set up reminders.
    - **Steps:**
        1. Wait for the scheduled time for the daily check-in reminder.
        2. Verify if the system sends a push notification or alert.

        - **Expected Result:** A notification should be sent to the user at the scheduled time.

5. **Professional Support Directory**
    - **Test Case 5.1:** Verify that the system correctly displays a list of mental health professionals when requested.
        - **Precondition:** User is on the "Professional Support" screen.
        - **Steps:**
            1. Select a category (e.g., therapist, counselor, helpline).
            2. Provide location-based filters (e.g., city, state).
        - **Expected Result:** A list of available professionals should be displayed based on the filters applied.
    - **Test Case 5.2:** Verify that emergency helpline contacts are displayed and accessible.
        - **Precondition:** User requests emergency support.
        - **Steps:**
            1. Navigate to the emergency helpline section.
            2. Click on the helpline number or link.
        - **Expected Result:** The helpline number should be displayed, or a link should initiate a phone call or message to the helpline.
6. **Error Handling and Edge Cases**
    - **Test Case 6.1:** Verify that the system handles incomplete or invalid user input during daily check-in.
        - **Precondition:** User enters incomplete or invalid data (e.g., leaving one field blank).
        - **Steps:**
            1. Attempt to submit the check-in with missing or invalid data.
        - **Expected Result:** System should display an error message indicating the invalid field (e.g., "Please fill out all fields").

- **Test Case 6.2:** Verify that the system gracefully handles network errors (e.g., if the server is unreachable).
    - **Precondition:** User is connected to the internet.
    - **Steps:**
        1. Disconnect from the internet or simulate a network error.
        2. Attempt to submit data or access feedback.
    - **Expected Result:** The system should show a user-friendly error message, e.g., "Network error. Please try again later."

7.3. Levels of Testing

Testing occurs at different levels, which enables us to methodically validate both modules, and the complete integrated system. Each testing level helps find bugs earlier in the development life cycle, and from each stage we can validate the system as a whole.

7.3.1. Unit Testing

Objective: Test each individual function or module, to ensure correct functionality in isolation.

Scope: Different functions/modules like user input validation, like the feedback that will be generated, possibly data processing.

Example:

- To ensure that the mood input function records and validates the correct mood value.
- To make sure that the stress level input function correctly records the user data, and processes it.
- To ensure that the recommendation algorithm is providing us with the correct coping strategies based on user input, like if the stress levels were high it would provide stress-relief exercise recommendations.

Tools: Jest, Mocha (for backend validation), JUnit (if using Java for backend).

7.3.2. Integration Testing

Objective: To ensure that the different components of the system (frontend, backend, database) effectively work together and do not interfere with one another.

Scope: Frontend interactions with backend (like the API calls), connections to the database.Example:

- Ensuring that the **daily check-in data** from the mobile app is correctly passed to the backend and stored in the database.
- Testing whether the **feedback generation** API correctly processes the data and sends tailored recommendations back to the user.

Tools: Postman (for API testing), Mocha, JUnit, and other testing frameworks.

7.3.3. System Testing

Objective: Test the entire system as a whole to ensure it works according to the specified requirements.

Scope: All features working together in the application, including the user interface, data processing, and integration with external resources.

Example:

- Testing the entire workflow of the app: user registration → daily check-in → feedback generation → progress tracking → accessing professional support resources.
- Verifying that the **notifications system** works as expected (e.g., daily reminders for check-ins).

Tools: Selenium (for automated UI testing), TestComplete.

7.3.4. Sanity Testing

Objective: Quickly verify that major functionalities are working after minor changes or bug fixes.

Scope: A quick round of testing for specific areas that were modified, ensuring they are functional.

Example:

- After fixing an issue with the **progress tracking feature**, a sanity test would confirm that the **mood tracking graph** now displays properly.
- Verifying that the **notification feature** works after changes to the reminder system.

Tools: Manual testing or basic automated testing scripts.

7.3.5. Smoke Testing

Objective: Perform a basic check to ensure that the application is stable enough for more extensive testing.

Scope: Critical path functionality (e.g., the app launches correctly, users can log in, and daily check-ins can be submitted).

Example:

- Ensuring the app opens, the user can log in, the check-in feature works, and that the user can view basic feedback.
- Verifying that the app doesn't crash on launch, and essential buttons and screens are functional.

Tools: Manual testing or simple automated tests.

7.3.6. Acceptance Testing

Objective: Verify that the app meets the requirements and is ready for release or deployment.

Scope: Ensure that all user stories and business requirements have been implemented and meet the acceptance criteria.

Example:

- Confirming that the app allows users to track mental health (mood, stress, symptoms) over time, provides personalized feedback, and integrates emergency professional support.
- Validating that the system fulfills the project requirements, such as correct progress tracking, user-friendly interfaces, and daily check-in reminders.

Tools: Cucumber (for behavior-driven development), manual testing based on user stories.

7.3.7. Regression Testing

Objective: Ensure that new changes (such as bug fixes or feature additions)

haven't introduced new issues or broken existing functionality.

Scope: Re-test major areas of the app that might have been affected by recent changes

(e.g., user registration, check-ins, notifications).

Example:

- After adding a new feature for **professional support contacts**, regression testing would check if the **feedback system** and **progress tracking** are still functioning properly.
- Verifying that **mood trend graphs** and **activity tracking** still work after updates to the system.

Tools: Selenium, JUnit, Mocha.

7.3.8. Performance Testing

Objective: Evaluate the performance of the app under load conditions to ensure

it works effectively with a large number of users.

Scope: Load and stress testing to evaluate the system's ability to handle multiple users.

Example:

- Verifying that the app can handle a high volume of users performing daily check-ins simultaneously without crashing or slowing down.
- Ensuring the app's response time remains fast (e.g., for generating feedback or displaying progress).

Tools: JMeter, LoadRunner, Apache Benchmark.

7.3.9. Security Testing

Objective: Identify vulnerabilities in the system, especially concerning sensitive data (e.g., user profiles, mental health data).

Scope: Testing for common security issues like unauthorized access, data leaks, and weak encryption.

Example:

- Ensuring that **user data (e.g., mental health information, personal details)** is stored securely with encryption.
- Verifying that the app does not allow **unauthorized access** to users' mental health data or personal information.
- Checking for vulnerabilities such as **SQL injection**, **cross-site scripting (XSS)**, and **insecure APIs**.

Tools: OWASP ZAP, Burp Suite, manual security audits.

7.3.10. User Interface (UI) Testing

Objective: Ensure the user interface is intuitive, responsive, and meets the design specifications.

Scope: Verifying the app's layout, design, and usability.

Example:

- Checking that **buttons**, **forms**, and **feedback sections** are visible and functional on different screen sizes.
- Testing for a consistent user experience across **mobile devices**, including landscape/portrait modes.
- Ensuring that **text labels** and **icons** are clearly visible and easily understood by users.

Tools: Selenium, TestComplete, or manual UI testing.

7.3.11. Usability Testing

Objective: Ensure that the app is user-friendly, especially for individuals with mental health concerns, and is easy to navigate.

Scope: Test how easily users can navigate through the app and access features such as check-ins, feedback, and professional support.

Example:

- Conducting tests with real users to see if they can easily complete tasks such as tracking their mood or accessing emergency support without confusion.
- Checking the **clarity** and **simplicity** of the **feedback system**, ensuring it is helpful and actionable.

Tools: Manual testing, A/B testing, user interviews.

7.4. Testing the Project

The extensive testing procedures carried out on the Mental Health Tracking and Support System confirmed that the mobile application meets its functional, security, and performance requirements. Functional testing established that the primary functions of mood tracking, feedback provision, and professional support alignment were working correctly. Usability testing showed that the mobile application was easy to navigate and user-friendly, while security testing confirmed the safeguarding of user data through encryption and access control.

Performance testing established that the application met user loads, and

compatibility testing confirmed that different devices and web browsers functioned correctly. Regression and smoke testing confirmed that updates and fixes did not impede core application functionality, and user acceptance testing established that the application met the needs of users through trial usage.

Finally, system and stress testing confirmed that all system parts and components worked in harmony, and the application could operate under extreme user loads. All components of the Mental Health Tracking and Support System have been effectively tested and confirmed their readiness for potential production deployment for users. With thorough testing processes, users can interact with the system in a secure, reliable, and positive way.

8. IMPLEMENTATION

8.1. Implementation of the Project

The Mental Health Tracking and Support System was implemented in stages, each of which focused on a specific task and core components needed to fulfill the required functionality. The various stages of implementation are listed below:

1. Requirement Analysis and Design

- The first phase of the implementation involved identifying and understanding the project requirements (e.g. mood formats, personalized feedback, professional support directory, and user profile).
- Based on these requirements, a system architecture was designed that included a user-friendly frontend, a secure backend, and a scalable database to store user data.
- The system was designed to ensure a smooth user experience, focusing on accessibility, data security, and ease of use.

2. Frontend Development

- The frontend of the app was developed using React (or your chosen framework), which allowed for a responsive and interactive interface.
- Mood tracking features were implemented through simple forms, where users could input their mood ratings, and these were then stored in the backend.
- The feedback generation system was integrated to provide users with personalized messages based on their mood logs and tracked patterns over time.
- A professional support directory was integrated into the frontend to allow users to access a list of available mental health professionals.

- The UI was built to ensure a smooth navigation flow, offering easy access to each feature and ensuring accessibility for all users.

3. Backend Development

- The backend was developed using Node.js with Express (or an appropriate backend technology) to manage user requests, authentication, and data processing.
- User authentication was implemented using hashed passwords, ensuring secure login and registration processes.
- The backend also handled mood data storage, integrating with the database (e.g., MySQL or MongoDB) to track and store mood logs and feedback generated for each user.
- An API was developed to allow seamless interaction between the frontend and backend for fetching mood data, submitting feedback, and accessing professional support information.

4. Database Implementation

- The database was designed to store user profiles, mood logs, feedback history, and mental health professional data.
- A MySQL or MongoDB database was chosen for scalability and efficient querying.
- Data relationships were carefully defined to ensure efficient retrieval of mood data, feedback, and user profiles.

5. Security Features

- Data encryption was implemented to protect sensitive user information, including mood logs and personal details.
- JWT (JSON Web Tokens) were used for secure authentication and authorization, ensuring only authorized users could access their personal data.
- Regular security checks were performed to ensure the app was protected against common vulnerabilities such as SQL injection, cross-site scripting (XSS), and data breaches.

6. Testing and Debugging

- After the initial implementation, unit testing and integration testing were carried out to ensure the individual components of the app were working as intended.

- **Bug fixes** and refinements were made based on the results from the testing phase, ensuring smooth and error-free functionality across the entire system.

7. Deployment

- Once the system was fully implemented and tested, it was deployed to a cloud platform such as **AWS** or **Heroku**, ensuring high availability and easy scalability.
- The frontend was hosted using **Netlify** or **Vercel**, and the backend API was hosted on **Heroku** or **AWS EC2**.
- The deployment process also involved setting up a **CI/CD pipeline** to automate updates and ensure smooth deployment of future changes.

8. Final User Testing and Feedback

- After deployment, **user acceptance testing (UAT)** was conducted with real users to gather feedback on usability and overall experience.
- Based on the feedback, final adjustments were made to improve user experience, fix any remaining issues, and add enhancements where necessary.

9. Documentation

- A comprehensive **user manual** and **developer documentation** were created, including system architecture diagrams, API documentation, and deployment instructions, to ensure future maintainability and ease of use.
- The documentation was structured in accordance with academic and project submission guidelines.

9. PROJECT LEGACY

The project for the Mental Health Tracking and Support System has generated a substantial legacy beyond its initial implementation. This legacy is demonstrated in its possible effects on users, the mental health sector, and the lasting positive effects for individuals and organizations that develop and use the Tracking System.

1. **Impact on Mental Health Support**

- Personalized Mental Health Monitoring The Monitoring System provides users with an exciting new resource to document their emotional wellbeing over time. By prompting regular mood check-ins and offering personalized recommendations, the app encourages users to take a more active role in managing their own mental health and wellbeing.

- Accessibility to Services. By incorporating professional support services into the platform, users have access to mental health professionals and mental health professional resources and productivity tools all in one location. This accessibility could have huge implications for individuals who have no access to mental health support due to their geographical location or stigma related to mental health support.

- Engagement with Data. The project promotes awareness of the ability to access, explore and analyze data for future discussion. The Tracking System promotes a user manual for mood to examine alignment of the user journey over time. By tracking mood trends over time, users may obtain insight into what influences their mood and ultimately are better positioned to make more thoughtful decisions and essentially practice better self-care.

2. Educational Legacy

- **Learning and Growth:** The project has created learning opportunities for everyone involved in developing/creating it. It has facilitated experiential education related to system architecture design, database architecture, security, and user-centered design. This unique knowledge will stay with the team and it can be used again for future projects.

- **Cross-Disciplinary Application:** The amalgamation of technology and mental health care, in the project contains links to both technical development and health care; it can lead to cross disciplinary linking and potentially new and interesting tech related innovations that respond to society's needs.

3. Technological Contributions

- **Scalable and Secure Architecture:** The system employs modern development practices (secure user authentication, API integration, and cloud-hosted/deployed architecture); this allows newly-built systems to be safe and scalable. Any limitations which may not make sense at first can usually be addressed through purely technical means. New ideas can be realized, or at least developed incrementally, based on the limitations or revisions, by having a secure and scalable infrastructure.

- **Innovative Features:** Features such as mood monitoring, personalized feedback, and access to mental health practitioners are examples of innovations that have made this project possible and which may, in some manner, contribute to more sophisticated mental health technology development in the future. Ultimately, there might be new developments in technology designed to address mental health applications in the future.

- **Data Privacy and Security:** The emphasis on data encryption and access controls establishes a sophisticated standard for future mental health apps, ensuring privacy is respected, and protected, which is especially important in the sensitive domain of mental health.

4. **Social and Cultural Legacy**

- **Reducing Mental Health Stigma:** By producing an easily available platform dedicated to mental health support, the system aims to normalize mental health care. Over time, these initiatives can help reduce stigma related to mental health and motivate individuals to pursue help without worry of being judged.

- **Promoting Mental Well-Being:** As more individuals utilize technology to manage their well-being, the project aims to create a culture of self-care and proactive management of mental well-being – one that could influence individuals and organizations to prioritize mental health in personal and professional instances.

5. **Future Growth and Adaptability**

- **Sustainability and Future Updates:** The system has been designed to be scalable, ensuring that future upgrades, if desired, and additional features can be easily incorporated into the system/project, if desired. Future updates could include further AI-based feedback capabilities, integration with wearable devices to track user data in real-time, or expanding the networks of mental health professionals, as examples.

- **Potential for Widespread Adoption:** Because the system is easy to use, and based on evidence and the system is operational, it could be easily rolled out at a larger scale and within larger organizations, such as universities, corporations, and NGOs focused on mental health, as part of their employee wellness programs or outreach initiatives.

10. USER MANUAL

1. Introduction

- The Mental Health Tracking and Support System is designed to help users track their mental well-being, receive personalized feedback, and access professional mental health support. This system empowers individuals to monitor their

emotional health and seek guidance from qualified professionals.

2. Getting Started

System Requirements

- **Device:** Smartphone, Tablet, or Computer
- **Operating System:** Android, iOS, Windows, MacOS
- **Internet Connection:** Required for account setup, professional support access, and feedback synchronization.
- **Browser Compatibility:** Google Chrome, Firefox, Safari, Microsoft Edge

Installation Guide

1. Visit the **official website** or **app store** to download the system.
2. Follow the on-screen instructions to install the app on your device.
3. Once installed, launch the app to start the setup process.

3. Creating an Account

Sign Up

1. Open the app and click on **"Sign Up"**.
2. Fill in the required fields:
    - **Username** (Your preferred user ID)
    - **Email address** (Used for account verification and notifications)
    - **Password** (Ensure it's strong and secure)
3. Click **"Create Account"** to submit your details.

Login

1. Open the app and click **"Login"**.
2. Enter your **Username** and **Password**.
3. Click **"Login"** to access your account.

Account Recovery

1. If you forget your password, click **"Forgot Password?"** on the login screen.
2. Enter your registered email address, and instructions will be sent to reset your password.
3. Follow the link in the email to set a new password.

4. Main Features

Mood Tracking

- **How to Track Mood:**

1. On the homepage, click the **"Track Mood"** button.
2. Rate your mood on a scale from 1 to 10 (1 being very low, 10 being very high).
3. Optionally, write a brief description of your mood or any related thoughts.
4. Click **"Save"** to log your mood.

Feedback Generation

- After tracking your mood, personalized feedback will be generated based on your input.
  - You will receive guidance on how to manage your emotional state, along with suggestions for self-care or professional support if needed.
  - **View Feedback:** Go to the **"My Feedback"** section to view past feedback based on your mood logs.

Professional Support Access

1. Navigate to the **"Professional Support"** section.
2. Browse through a list of licensed mental health professionals.
3. Select a professional based on your needs and schedule an appointment or reach out via chat or call.

Personal Profile Management

- Access your profile settings from the **"Profile"** section to:
  - Update personal details (name, email).
  - Change your password.
  - Manage notification preferences.

5. Using the System

How to Track Mood

1. On the homepage, click **"Track Mood"**.
2. Choose your mood rating and add any additional notes.
3. Save your mood log to receive personalized feedback.

How to View Feedback

1. Click on **"My Feedback"** from the main menu.
2. Browse through past feedback entries, which will be automatically generated based on your mood logs.
3. Read suggested actions and tips for improving your emotional well-being.

How to Access Professional Support

1. Go to **"Professional Support"**.
2. Browse the list of professionals by specialization (e.g., anxiety, depression).
3. Click on a professional's name to see their profile and book an appointment.

Managing Your Profile

1. Navigate to **"Profile"**.
2. Click **"Edit Profile"** to update personal information.
3. Adjust notification preferences to control the types of alerts you receive.

6. Security and Privacy

Data Encryption

- All user data, including mood logs, feedback, and personal information, is encrypted using industry-standard security protocols to ensure your privacy and data protection.

User Authentication

- Secure login is enabled using hashed passwords and **two-factor authentication** (optional) to protect user accounts.

Privacy Policy

- We take user privacy seriously. For more details on how we protect your information, please read our Privacy Policy.

7. Troubleshooting

- **App Crashing:** If the app crashes, try restarting your device or reinstalling the app.
- **Login Issues:** Ensure your username and password are correct. If issues persist, use the **Forgot Password** feature to reset your password.
- **Mood Logging Problems:** If mood logs are not saving, check your internet connection or try restarting the app.
- Browse the list of professionals by specialization (e.g., anxiety, depression).
- Click on a professional's name to see their profile and book an appointment.

Managing Your Profile

- Navigate to **"Profile"**.
- Click **"Edit Profile"** to update personal information.
- Adjust notification preferences to control the types of alerts you receive.

Working Project:

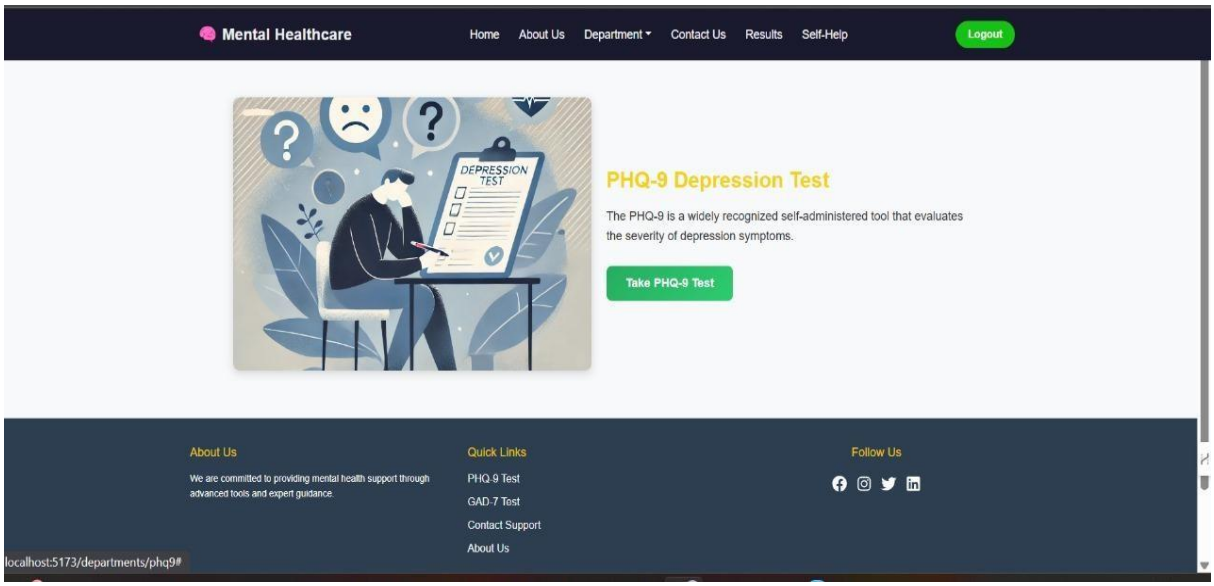


Fig- 4 PHQ-9 Test Inteface

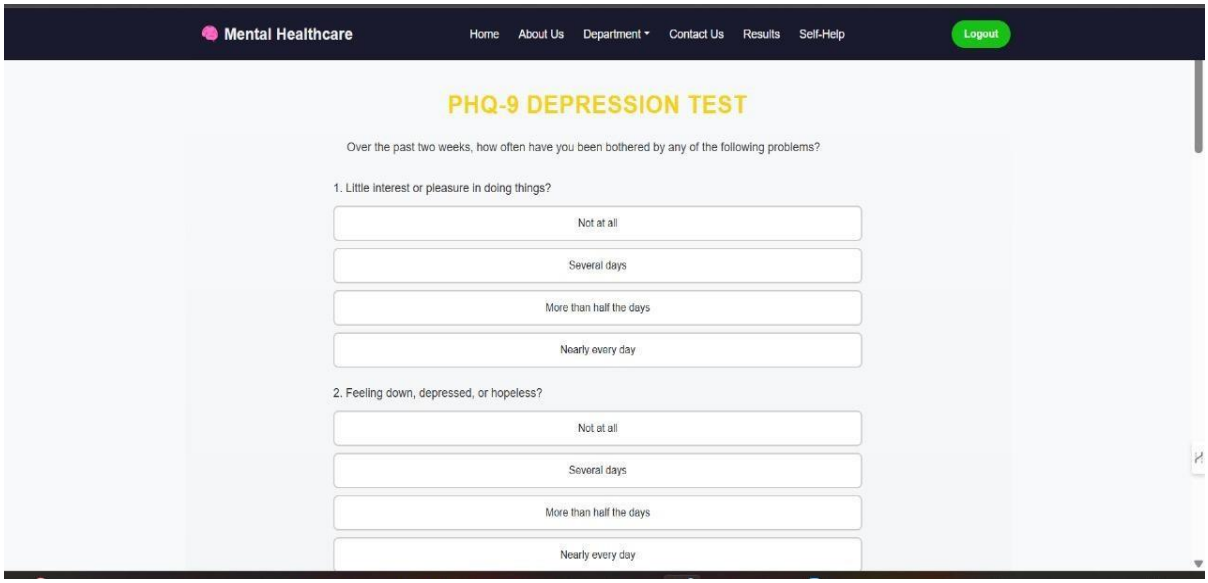


Fig- 5 PHQ-9 Questionaire

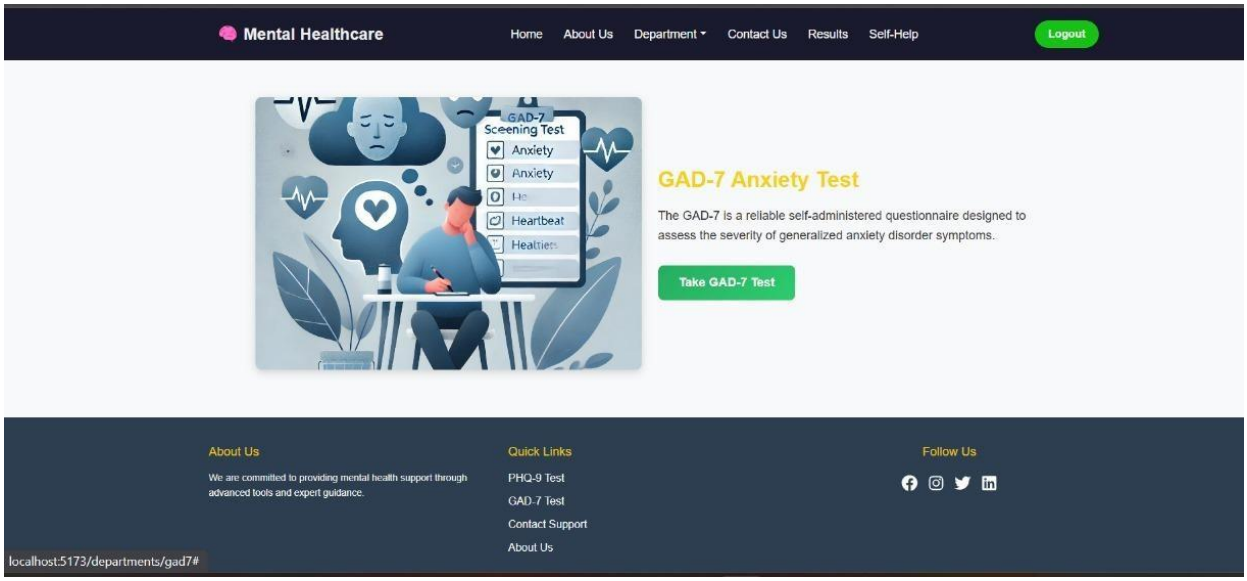


Fig -6 GAD-7 Test Interface

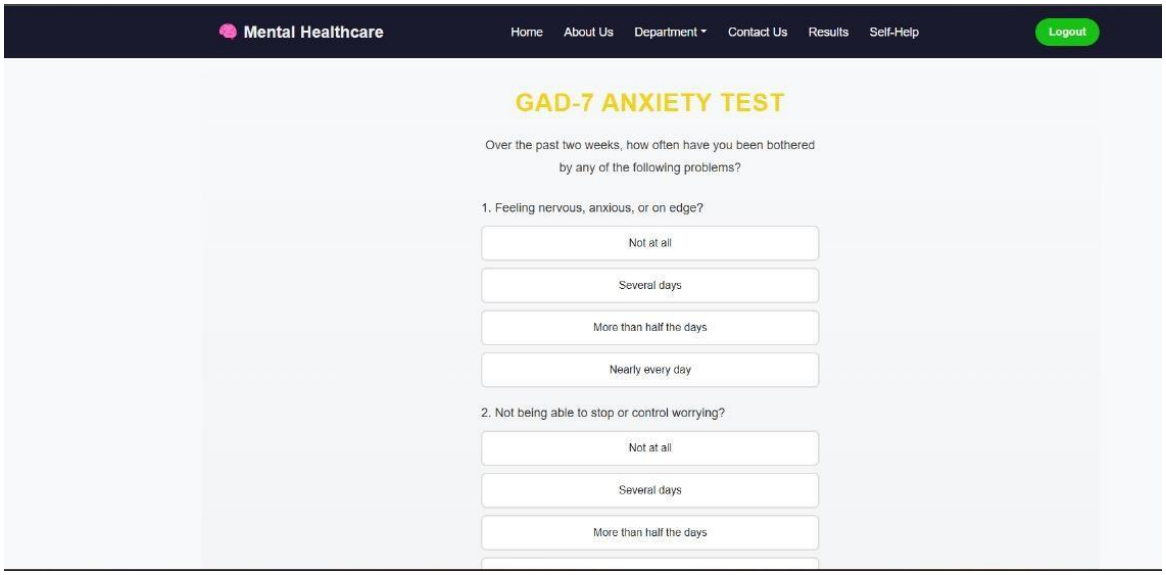
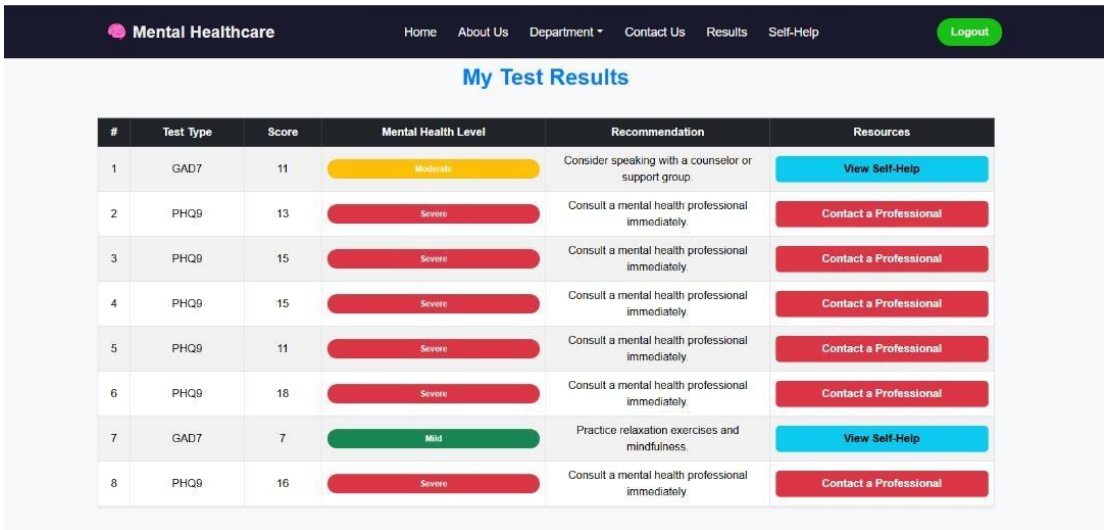


Fig -7 GAD-7 Questionaire

Results:



#	Test Type	Score	Mental Health Level	Recommendation	Resources
1	GAD7	11	Moderate	Consider speaking with a counselor or support group.	View Self-Help
2	PHQ9	13	Severe	Consult a mental health professional immediately.	Contact a Professional
3	PHQ9	15	Severe	Consult a mental health professional immediately.	Contact a Professional
4	PHQ9	15	Severe	Consult a mental health professional immediately.	Contact a Professional
5	PHQ9	11	Severe	Consult a mental health professional immediately.	Contact a Professional
6	PHQ9	18	Severe	Consult a mental health professional immediately.	Contact a Professional
7	GAD7	7	Mild	Practice relaxation exercises and mindfulness.	View Self-Help
8	PHQ9	16	Severe	Consult a mental health professional immediately.	Contact a Professional

Fig- 8 Results of PHQ-9 and GAD-7

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